

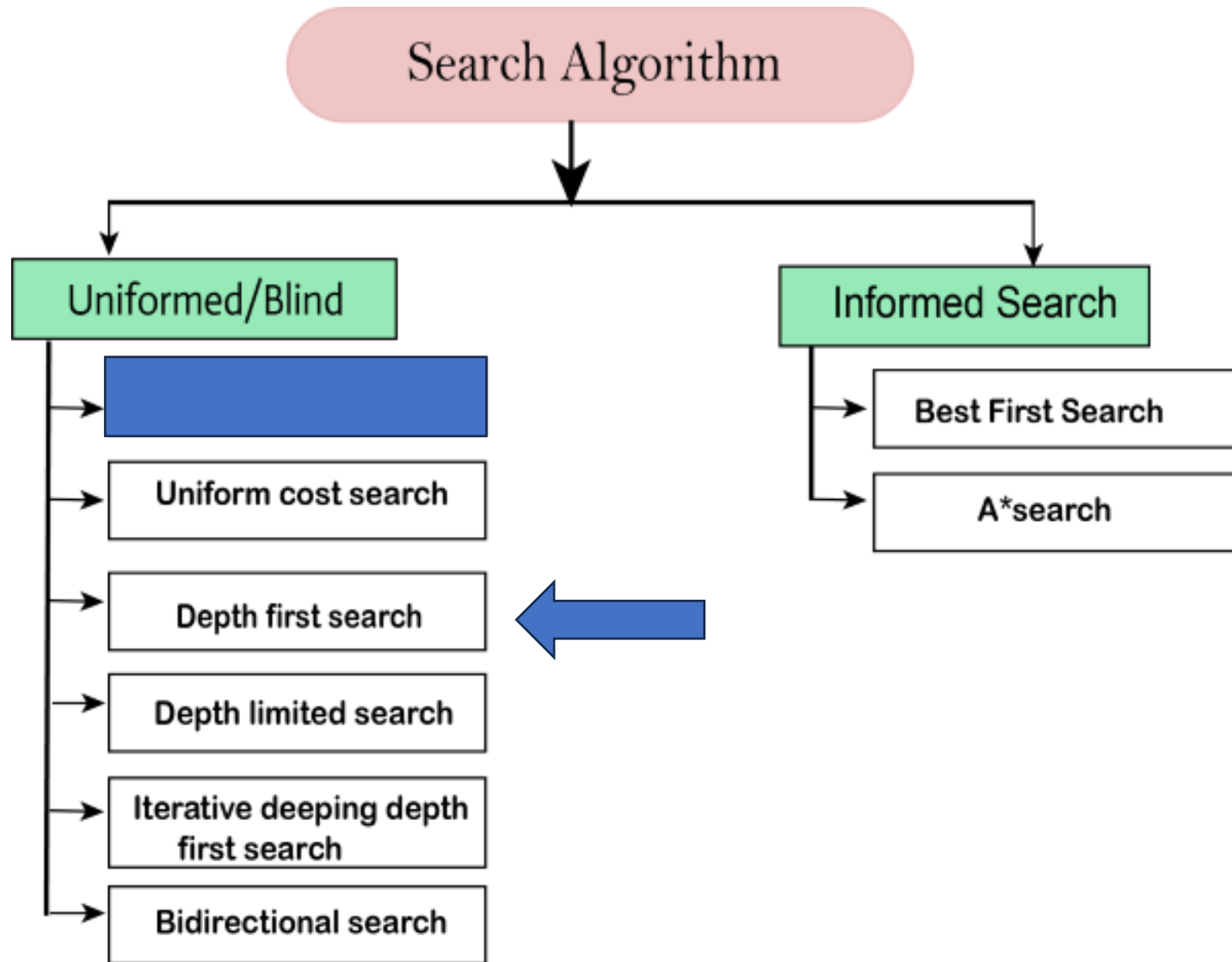
Artificial Intelligence

Section 5

By: Mohamed Fathy

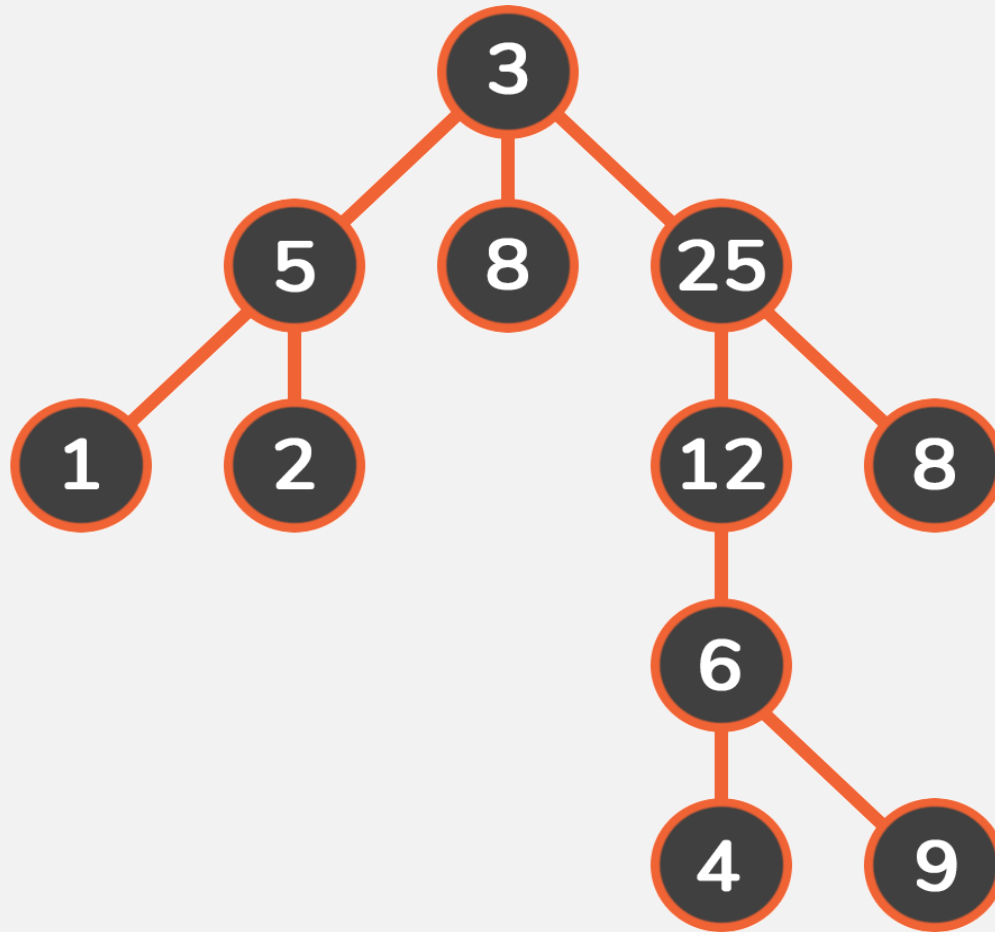
Section Content

- Depth-first Search
- Solving Romania Problem DFS
- Solving 8-Puzzel Problem using DFS



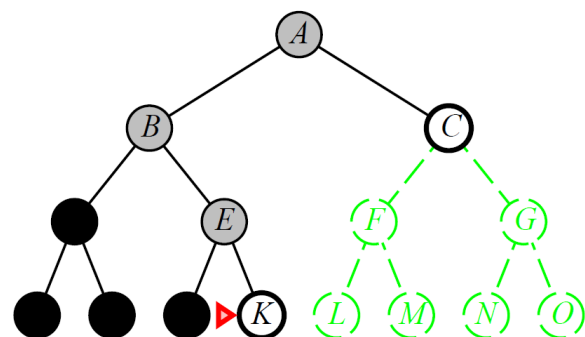
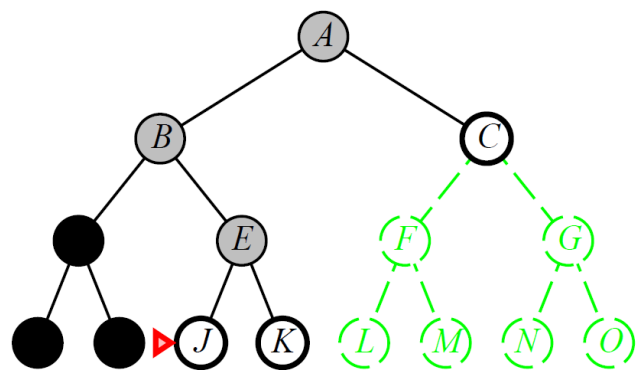
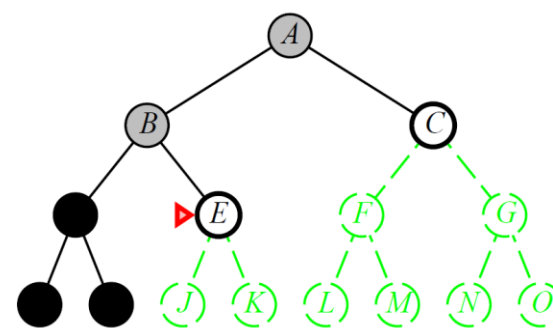
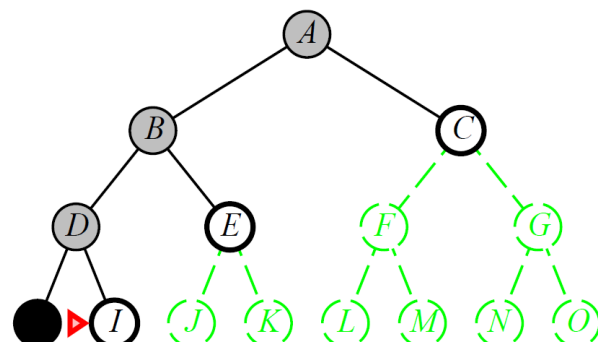
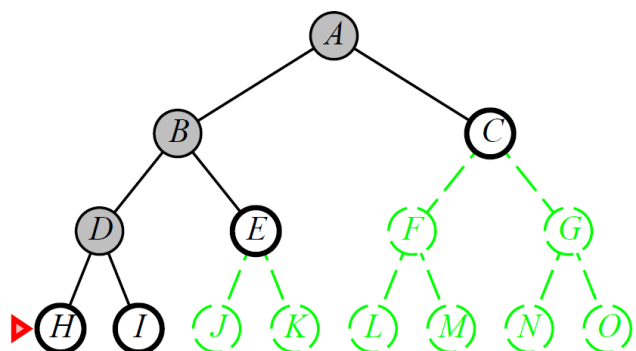
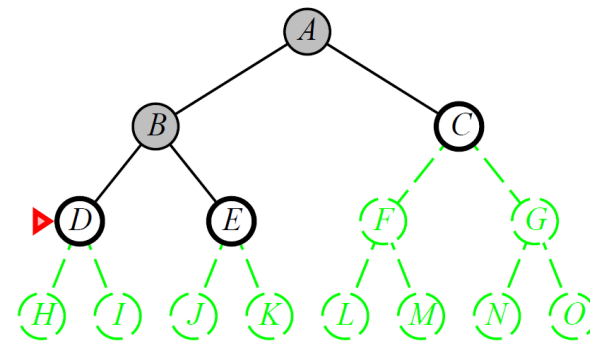
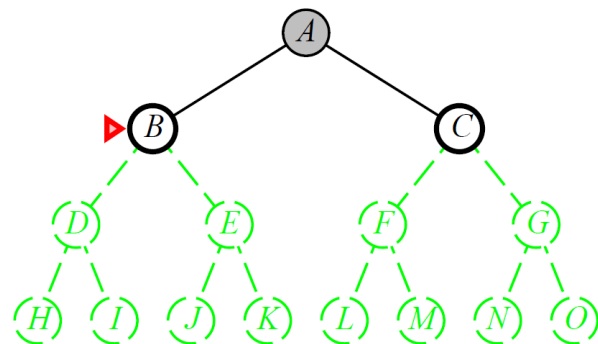
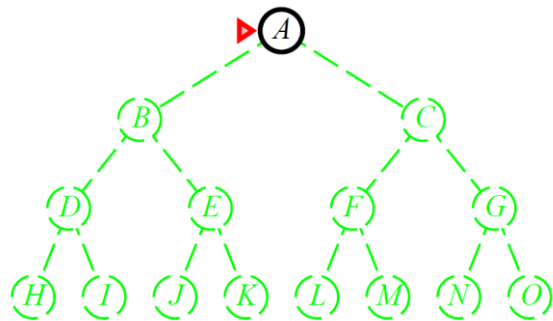
**Depth-first
Search**

- Definition
 - Depth-first search is a recursive algorithm for traversing a tree or graph data structure.
 - It is called the depth-first search because it starts from the root node and follows each path to its greatest depth node before moving to the next path.
- Advantages
 - DFS requires very less memory as it only needs to store a stack of the nodes on the path from root node to the current node.
 - It takes less time to reach to the goal node than BFS algorithm (**if it traverses in the right path**).
- Disadvantages
 - There is the possibility that many states keep re-occurring, and there is no guarantee of finding the solution.
 - DFS algorithm goes for deep down searching and sometimes it may go to the infinite loop.



Depth-first Search

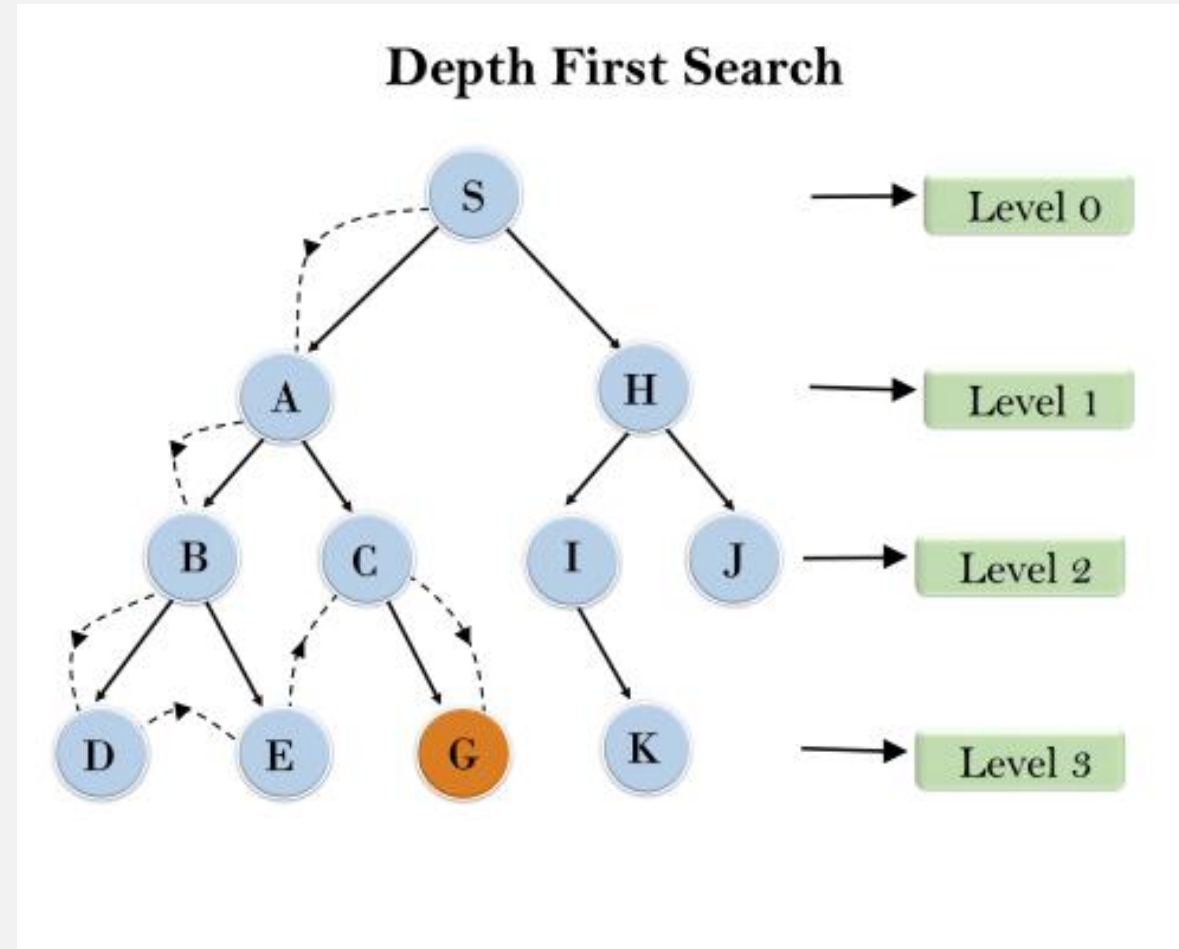
The process of the DFS algorithm is similar to the BFS algorithm.



**Depth-first
Search**

- In the below search tree, we have shown the flow of depth-first search, and it will follow the order as:

S---> A--->B---->D--->E---->C--->G



**Depth-first
Search**

- The process of the DFS algorithm is similar to the BFS algorithm.
- DFS uses a **stack** data structure for its implementation.
- A variant of depth-first search called **backtracking search** uses even less memory.

Romania Problem Implementation (DFS)

Depth-first
Search

Solving 8-Puzzle Problem

Depth-first
Search

- **Time Complexity:** Time complexity of DFS will be equivalent to the node traversed by the algorithm. It is given by:

$$T(n) = 1 + n^2 + n^3 + \dots + n^m = O(n^m)$$

Where, m = maximum depth of any node and this can be much larger than d (Shallowest solution depth)

- **Space Complexity:** DFS algorithm needs to store only single path from the root node, hence space complexity of DFS is equivalent to the size of the fringe set, which is **$O(bm)$** .
- **Completeness:** DFS search algorithm is **complete** within finite state space as it will expand every node within a limited search tree. In infinite state spaces, depth-first search is not systematic: it can get stuck going down an infinite path, even if there are no cycles. Thus, depth-first search is **incomplete**.
- **Optimality:** DFS search algorithm is **non-optimal**, as it may generate a large number of steps or high cost to reach to the goal node.

**Depth-first
Search**